TRIBUNAL · Day Pack

Build with Gemini 4 Argon — Imperial College × Google DeepMind hackathon · Track 1: Frontier Agents · Oskar · as of 31 Oct 2026

1 · Download & setup (laptop, PowerShell)

The file: **`imperial-gemini-tribunal-2026-10-31.tar.gz`** in the transfers bucket (1.6 MB — full app, battle plan, demo docs, golden replay traces, vendored Gemini API docs; only the venv and the cookbook repo are excluded). Python 3.11+ (3.12 preferred).

```
cd $HOME\Downloads
aws s3 cp
"s3://celestia-data-730335578848/transfers/imperial-gemini-tribunal-2026-10-31.tar.gz" .
--region eu-west-2
tar -xzf imperial-gemini-tribunal-2026-10-31.tar.gz
cd imperial-gemini
python -m venv .venv
.venv\Scripts\pip install -r requirements.txt
.venv\Scripts\python -m tribunal.server --mock # then open http://localhost:8990
```

That is the whole install. No API key needed until kickoff — everything runs in mock mode.

2 · Tonight's rehearsal, then kickoff

Five minutes tonight: paste any question → Enter → click a row in the approval modal to **strike** it → Approve → in the jury panel click **Rule for this source** → watch the verdict stream in. Then press **R** — the instant-replay fallback hotkey. Get both into muscle memory. Sanity check: `.venv\Scripts\python scripts\e2e_mock.py` → PASS.

At kickoff, key in hand: (1) `$env:GEMINI_API_KEY="<key>"` · (2) work through **docs\demo\kickoff-checklist.md** top to bottom — first live smoke test, the 2/4/8/14 grounding-concurrency ramp, every assumption only a live key can verify · (3) follow **TRIBUNAL-PLAN.md §6** hour by hour; CloseLoop pivot gate at H3.5. Prompts in tribunal\orchestrator\prompts\ are re-loaded every run — calibrate by editing the files.

3 · What Tribunal is — the one-breath pitch

> *"Tribunal is a research court. Gemini 4 Argon is the Judge: it breaks a contested technical question into a claim board, then parallel Flash Advocates argue each claim with live Google Search evidence while Skeptic agents try to kill every finding. What the agents can't settle goes to a human jury — me, or you. Out the end comes a cited verdict I can spend real money against."*

The Track 1 brief says "autonomous agents, multi-step reasoning, verification loops, research workflows" — Tribunal is that sentence performed on stage. The demo question is your real pre-spend due diligence: the cubic boron arsenide thermal-conductivity controversy, gating a NEO GPU-screening decision.

4 · The four judging tests — and how Tribunal passes each

Official test	How Tribunal passes it
1 · A specific problem for a real, nameable person	The named user is **you**: before committing NEO's paid GPU/DFT screening runs you must verify contested literature numbers — and a wrong one already cost you a GPU-weekend. You're standing in the room; no invented persona.
2 · Work that genuinely needed Gemini 4 Argon	Multi-agent orchestration (Argon Judge + 8–10 search-grounded gemini-3.8-flash litigants + flash-lite clerk); a **visible** verification loop (Skeptics exist solely to refute Advocates, in red, on screen); 1M-window synthesis holding every leaf transcript and quote — no RAG — plus mid-run re-planning on the human ruling via previous_interaction_id. A chat wrapper can do none of these.
3 · Humans stay in control	Two structural gates: the approval modal where a judge personally **strikes a plan branch** (engine-enforced — it never spawns and is provably absent from the brief), and the **jury** — a contested claim cannot resolve without a human ruling, stamped "human-adjudicated" forever.

Official test	How Tribunal passes it
4 · Runs live, real inputs, under 3 minutes	Script cut to 2:30 with 30s slack. Provably live: a judge's own click kills a branch at 0:20, Skeptics pull real 2025 search results at 1:25, and you offer to run a judge's own question in Q&A. Four fallback layers degrade instead of dying: slow agents excluded → phone hotspot → cached-plan seed → hotkey-R replay, honestly badged REPLAY.

5 · Why it beats the room

- **Wrapper-proof by construction.** DeepMind judges are allergic to thin demos; an adversarial agent swarm with engine-enforced gates and an event-sourced trace is the opposite of a prompt in a textbox.
- **Legible at 3 metres.** Ten agents igniting at once, a full-screen DISPUTED takeover with two primary-source quotes, a gold HUMAN RULED seal — the room reacts without reading fine print.
- **Honesty as armor.** Provenance chips on every finding, "estimate" on the cost meter, a LIVE/REPLAY chip that never lies — every skeptical question has a visible answer already on screen.
- **Nothing rented.** Every runtime component is yours; the day's risks (rate limits, latency) are measurable in the first 30 minutes, with a pre-agreed pivot (CloseLoop, gate at H3.5).

6 · Numbers to know cold (the demo question)

Year	Value (W/mK)	Source	What it means
2013	~2200	Lindsay, Broido & Reinecke, PRL (theory)	First-principles three-phonon prediction
2015–17	~190–350	Lv et al., APL + others	Defect-limited early crystals
2017	~1400	Feng, Lindsay & Ruan, PRB (theory)	Four-phonon scattering cut the ceiling
2018	~1000–1300	Kang / Li / Tian — the *Science* trio	The famous headline measurements
2025	**>2100**	Niyikiza et al., *Materials Today*	**Arsenic-purified** crystals: impurities, not physics, were the limit

The 2025 number is **HIGHER** — the field's best value nearly doubled, so anyone designing against the 2018 papers is ~60% low. **Never say "isotope-pure"**: the isotope effect is only 8–13%; it was chemical purification. Full timeline, DOIs and ten hard-question answers: docs\demo\bas-controversy-armor.md.

7 · Q&A one-liners to have loaded

- **"Isn't this just Deep Research in a courtroom skin?"** — "Deep Research confirms; Tribunal cross-examines. Every claim gets an adversary, a contested claim can't resolve without a human, and the Judge re-plans mid-run on that ruling. Deep Research is literally a background component here."
- **"Run it on MY question."** — "Gladly — that's the second run. If yours turns out uncontested, the all-green board *is* the correct answer. And you already proved nothing's canned: you struck a branch and it never ran."
- **"How do I know this was live?"** — Point at the branch they struck, absent from the brief; then at the trace file, and the REPLAY chip that would have been showing if it weren't. All six hardest questions with full answers: TRIBUNAL-PLAN.md §11.

8 · Print / read before you go

- [ ] Print **docs\demo\judge-card.md** — hand it to a judge (strike, approve, rule)
- [ ] Print **docs\demo\beat-sheet.md** — your 2:30 script with the fallback lines
- [ ] Memorize **docs\demo\bas-controversy-armor.md**
- [ ] Read **TRIBUNAL-PLAN.md** §§6–8 and §11 once, end to end
- [ ] Pair the phone hotspot with the laptop (fallback layer 2)